

ORIGINAL ARTICLE

Optimal Strain Combinations for Prognostic Prediction in Pulmonary Arterial Hypertension

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BACKGROUND: Strain parameters are sensitive indicators of myocardial performance, but the optimal combination for prognostic prediction in pulmonary arterial hypertension remains unclear.

METHODS: Patients diagnosed with pulmonary arterial hypertension between June 2013 and June 2024 were prospectively recruited. Strain parameters, including left ventricular global longitudinal strain, right ventricular GLS, right ventricular free-wall longitudinal strain, and right atrial phasic strain, were analyzed. Feature selection used elastic net Cox regression and stepwise Cox regression with the Akaike information criterion. Survival analyses were performed for all-cause mortality and a composite end point (mortality and heart failure rehospitalization).

RESULTS: A total of 269 patients with pulmonary arterial hypertension (mean age, 42±15 years) were included. During a median follow-up of 39.3 months, 68 patients reached the primary end point, and 91 reached the secondary end point. The combination of left ventricular global longitudinal strain and right atrial reservoir strain had the strongest prognostic value. Both left ventricular global longitudinal strain (hazard ratio, 1.46, per 5% increase [95% CI, 1.07–2.00]; $P=0.02$) and right atrial reservoir strain (hazard ratio, 0.77, per 5% increase [95% CI, 0.62–0.95]; $P=0.02$) remained independent predictors of all-cause mortality. Their combination outperformed conventional cardiac magnetic resonance and clinical models, as evidenced by higher C-index values (all $P\leq 0.05$), and stratified patients into low-, intermediate-, and high-risk groups with distinct 1-year mortality rates (2.8%, 6.2%, 21.0%, respectively; $P<0.001$). Landmark analysis confirmed their long-term prognostic relevance.

CONCLUSIONS: The combined assessment of left ventricular global longitudinal strain and right atrial reservoir strain yields the strongest prognostic utility among the strain parameters in pulmonary arterial hypertension, significantly improves risk stratification beyond several established cardiac magnetic resonance and clinical predictors, and holds promise as a practical tool for guiding management strategies.

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Key Words: global longitudinal strain ■ magnetic resonance spectroscopy ■ prognosis ■ pulmonary arterial hypertension ■ survival analysis

Pulmonary arterial hypertension (PAH) is a progressive disorder characterized by pulmonary vascular remodeling, increased right ventricular (RV) afterload, and eventual right heart failure and death.¹ Despite therapeutic advances, prognosis remains poor.² Accurate assessment of right heart function is

essential for both prognostic evaluation and risk stratification in PAH.

Cardiac magnetic resonance (CMR) is the reference standard for biventricular assessment and provides robust quantification of RV morphology and function. Traditional volumetric indices, including RV ejection fraction (RVEF),

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WHAT IS NEW?

- Among all strain parameters, the combination of left ventricular global longitudinal strain and right atrial reservoir strain demonstrated the highest prognostic value in patients with pulmonary arterial hypertension.
- Integrating left ventricular global longitudinal strain and right atrial reservoir strain into the risk assessment significantly outperformed the European Society of Cardiology/European Respiratory Society cardiac magnetic resonance model and improved the prognostic performance of conventional clinical models.

WHAT ARE THE CLINICAL IMPLICATIONS?

- The combination of left ventricular global longitudinal strain and right atrial reservoir strain maintained robust discriminatory capacity for both short- and long-term outcomes, supporting its value across different follow-up horizons.
- The combined assessment of left ventricular global longitudinal strain and right atrial reservoir strain may provide a concise and practical imaging-based approach for more refined risk stratification in pulmonary arterial hypertension.
- Incorporating this strain combination into routine cardiac magnetic resonance assessment may help identify high-risk patients who require closer follow-up, intensified monitoring, or more proactive therapeutic evaluation.

RV end-systolic volume index (RVESVi), and stroke volume index (SVi), have well-established prognostic value and form the basis of current guideline-recommended imaging models.^{3,4} However, these parameters may fail to capture early myocardial dysfunction, highlighting the need for more sensitive imaging markers. CMR-derived strain analysis has emerged as a powerful tool for detecting subclinical dysfunction. Left ventricular global longitudinal strain (LV GLS) has been associated with outcomes in PAH, even when LV ejection fraction is preserved.^{5,6} RV global longitudinal strain (RV GLS) and RV free-wall longitudinal strain (RV FWLS) provide incremental prognostic information beyond traditional RV indices, with RV FWLS, in particular, identified as a robust predictor of mortality and clinical worsening.^{7–9} In addition, right atrial (RA) phasic strain was independently associated with adverse outcomes.^{10,11} Importantly, recent invasive and imaging studies suggest that RA phasic strain reflects RV diastolic stiffness and impaired ventricular filling rather than isolated atrial contractile dysfunction, thereby serving as an early marker of right heart impairment.¹²

Although these studies have established the importance of strain parameters individually, a key knowledge gap persists: whether the integration of strain indices yields superior prognostic accuracy compared with

Nonstandard Abbreviations and Acronyms

AIC	Akaike information criterion
CMR	cardiac magnetic resonance
ESC/ERS	European Society of Cardiology/ European Respiratory Society
HR	hazard ratio
LV GLS	left ventricular global longitudinal strain
NT-proBNP	N-terminal pro-B-type natriuretic peptide
PAH	pulmonary arterial hypertension
PH	pulmonary hypertension
RA	right atrial
RV	right ventricular
RVEF	right ventricular ejection fraction
RVESVi	right ventricular end-systolic volume index
RV FWLS	right ventricular free-wall longitudinal strain
RV GLS	right ventricular global longitudinal strain
SVi	stroke volume index
WHO FC	World Health Organization functional class

single-parameter models remains uncertain. Furthermore, the optimal combination of strain parameters for risk stratification in PAH has not been defined. Thus, this study aimed to identify the most effective combination of strain parameters for prognostication of patients with PAH.

METHODS

Data Availability

The data, analytic methods, and study materials that support the findings of this study are available from the corresponding author on reasonable request.

Ethics Statement

This study was approved by the ethics committee of West China Hospital and the SingHealth Centralized Institutional Review Board. All participants underwent the same CMR and right heart catheterization protocols and provided written informed consent.

Participants diagnosed with PAH from June 2013 to June 2024 at West China Hospital and the National Heart Center Singapore were prospectively and continuously recruited. PAH was diagnosed according to the European Society of Cardiology/European Respiratory Society (ESC/ERS) guidelines.^{3,13} All patients underwent both right heart catheterization and CMR within a 72-hour interval. The inclusion criteria were: (1) patients with idiopathic PAH, connective

tissue disease–associated PAH, congenital heart disease–associated PAH with a closed shunt >1 year, or other PAH; and (2) patients aged >18 years. The exclusion criteria were: (1) patients with poor CMR image quality or with real-time cine; and (2) patients who were lost to follow-up. Patients with pulmonary hypertension (PH) due to left heart disease (group 2 PH) were excluded, defined as those with pulmonary capillary wedge pressure >15 mmHg or evidence of coexisting left-sided heart disease. Patients with PH due to lung diseases and hypoxia (group 3 PH) were excluded based on clinical evaluation, pulmonary function testing, and chest imaging findings. Patients with chronic thromboembolic PH (group 4 PH) were also excluded based on clinical history and imaging findings. In addition, patients with group 5 PH were excluded based on the underlying cause, including complex congenital heart disease, hematologic disorders, metabolic diseases, fibrosing mediastinitis, or other rare systemic conditions.

Right Heart Catheterization

Right heart catheterization was performed in a standardized manner using a 5F transfemoral catheter under local anesthesia.¹⁴ Cardiac output was calculated using the indirect Fick method, and the cardiac index was calculated as cardiac output indexed to the body surface area. Pulmonary vascular resistance was calculated using the following formula: pulmonary vascular resistance=[mean pulmonary arterial pressure–pulmonary capillary wedge pressure]/cardiac output.

Cardiac Magnetic Resonance

CMR imaging was conducted on a 3.0-T scanner (Magnetom Tim Trio; Siemens Healthineers, Erlangen, Germany) using a 32-channel phased-array cardiac coil, with breath-holding and electrocardiography gating. Cine images were obtained with steady-state free precession along consecutive short-axis slices from base to apex, and 2-, 3-, and 4-chamber long-axis views, following a standardized protocol.¹⁵ The scan parameters were the same as described previously.¹⁶

Biventricular volumes, ejection fraction, and mass were analyzed using dedicated software (QMass version 8.1; Medis Medical Imaging Systems, Leiden, the Netherlands) according to the standardized protocol of the Society of Cardiovascular Magnetic Resonance postprocessing guideline.¹⁷ Biventricular functional parameters were analyzed by meticulously delineating the endocardial and epicardial contours of the myocardium, including the papillary muscles and trabeculations and excluding epicardial fat. Biventricular mass and volumes were indexed to the body surface area. RV longitudinal strain was analyzed using CMR feature-tracking software (Medis Suite QStrain 2.0, Leiden, the Netherlands). RA longitudinal strain was assessed using fast long-axis strain analysis, as previously described.¹⁰ RA reservoir strain reflects atrial filling and compliance during ventricular systole, RA conduit strain represents passive emptying during early ventricular diastole, and RA booster strain reflects active atrial contraction during late diastole. More details are presented in the [Supplemental Material](#), with illustrative examples shown in [Figure S1](#). The CMR score was calculated as the average of the scores for RVEF, RVESVi, and RV SVi, using the thresholds from the 2022 ESC/ERS guideline,³

with adjustment based on the study by Celant et al.^{3,18} RVEF thresholds of 45% and 19% were used to define low-, intermediate-, and high-risk categories.

Study End Points

The primary end point of this study was all-cause mortality. The composite end point was a combination of heart failure–related readmission and all-cause mortality. Outcome events were ascertained through review of medical records and follow-up contact.

Statistical Analysis

The normality of variables was assessed using the Kolmogorov-Smirnov test. Continuous variables were expressed as mean±SD or median with interquartile range, and categorical variables as counts and percentages. The independent-samples *t* test and Mann-Whitney *U* test were used to compare continuous variables, whereas the χ^2 test was used for categorical variables. NT-proBNP (N-terminal pro-B-type natriuretic peptide) was log-transformed before analysis. Variables associated with all-cause mortality and HF hospitalization were identified using univariate and multivariate Cox analysis. To reduce multicollinearity and identify the most relevant prognostic features, elastic net penalized Cox regression was applied for the primary and composite end points.¹⁹ The optimal penalty parameter (λ) was determined using 10-fold cross-validation, and the λ_{1se} criterion was used to select the final model. Predictors with nonzero coefficients were considered significant contributors. Additional model comparisons were performed using stepwise Cox regression based on the Akaike information criterion (AIC), with a change in AIC (Δ AIC) ≥ 2 indicating meaningful improvement. Receiver operating characteristic curves were used to compare prognostic models, and optimal strain cutoffs were determined using the Youden index. Heatmaps illustrated risk distribution across parameter combinations. A 2-tailed $P < 0.05$ was considered statistically significant. Analyses were performed using SPSS 26.0 (IBM, Armonk, NY) and R 4.0.1 (Vienna, Austria).

RESULTS

Baseline Characteristics

A total of 269 patients with PAH were included. More details are presented in [Figure S2](#). The mean age was 42±15 years, and 58 (22%) were male. The cohort comprised 140 (52%) with idiopathic PAH, 79 (29%) with connective tissue disease–associated PAH, 35 (13%) with shunt-corrected congenital heart disease–associated PAH, and 15 (6%) with other PAH. The most frequent World Health Organization functional class (WHO FC) was class II (49%). Hemodynamic parameters included a mean pulmonary arterial pressure of 49±18 mmHg, cardiac index of 2.2±0.8 L/min per m², and pulmonary vascular resistance of 14±9 Wood units. The majority of patients (54%) were initiated on oral combination therapy. Baseline characteristics are summarized in Table 1.

Table 1. Baseline Characteristics

	Overall (N=269)	Survive (n=201)	Deceased (n=68)	P value
Age, y	42±15	42±14	43±17	0.42
Male, n (%)	58 (22)	40 (20)	18 (38)	0.20
SBP, mmHg	117±18	118±17	114±18	0.12
DBP, mmHg	75±14	75±14	75±13	0.47
6MWD, m	381±119	400±114	333±125	<0.001
NT-proBNP, ng/L	543 (157–1637)	387 (124–1011)	2051 (954–3499)	<0.001
WHO FC, n (%)				
I	63 (23)	55 (27)	8 (12)	<0.001*
II	133 (49)	99 (49)	34 (50)	
III	67 (25)	47 (24)	20 (29)	
IV	6 (2)	0 (0)	6 (9)	
PAH classification, n (%)				
I/HPAH	140 (52)	104 (52)	36 (53)	0.47*
PAH-CTD	79 (29)	56 (28)	23 (34)	
PAH-CHD	35 (13)	28 (14)	7 (10)	
Other PAH	15 (6)	13 (6)	2 (3)	
Hemodynamics				
mRAP, mmHg	8.7±5.2	8.4±5.0	9.6±5.6	0.27
mPAP, mmHg	49±18	47±17	58±18	<0.001
CI, L/min per m ²	2.2±0.8	2.2±0.8	2.0±0.6	0.06
SvO ₂ , %	65±10	66±10	62±10	0.20
PVR, Wood unit	14±9	13±9	17±10	0.002
PCWP, mmHg	9.3±4.3	9.3±4.6	9.3±4.2	0.94
Medications, n (%)				
Nil-targeted	15 (6)	14 (7)	1 (2)	0.26
Monotherapy	108 (40)	79 (39)	29 (43)	0.63
ERA	94 (35)	69 (34)	25 (37)	0.72
PDE5i	11 (4)	8 (4)	3 (4)	0.88
sGC	1 (0.4)	1 (0.5)	0 (0)	0.56
Dual combination	125 (46)	95 (47)	30 (44)	0.65
ERA+PDE5i	108 (40)	84 (42)	24 (35)	0.35
ERA+sGC	17 (6)	11 (5)	6 (9)	0.33
Triple combination	21 (8)	13 (7)	8 (12)	0.16
ERA+PDE5i+prostanoid	17 (6)	10 (5)	7 (10)	0.12
ERA+sGC+prostanoid	4 (2)	3 (2)	1 (2)	0.99
Cardiac MRI parameters				
LVEDVi, mL/m ²	63±22	64±18	62±31	0.41
LVESVi, mL/m ²	26±13	25±12	27±17	0.11
LV SVi, mL/m ²	38±13	39±12	35±17	0.06
LVEF, %	60±12	60±12	57±12	0.02
LV massi, g/m ²	42±14	42±12	42±20	0.34
RVEDVi, mL/m ²	116±53	108±47	139±64	<0.001
RVESVi, mL/m ²	74±45	67±40	98±50	<0.001
RVEF, %	38±15	41±15	31±12	<0.001
RV massi, g/m ²	28±12	26±11	34±12	<0.001
LV GLS, %	−19.7±4.4	−20.7±3.7	−17.0±5.2	<0.001
RV GLS, %	−19.2±7.6	−20.3±7.5	−15.8±6.9	<0.001

(Continued)

Table 1. Continued

	Overall (N=269)	Survive (n=201)	Deceased (n=68)	P value
RV FWLS, %	−20.9±9.0	−22.3±8.6	−16.6±8.3	<0.001
RA reservoir strain, %	27.3±12.7	30.1±11.5	20.5±9.9	<0.001
RA conduit strain, %	10.7±7.2	11.7±7.4	7.6±4.6	<0.001
RA booster strain, %	16.5±8.3	17.8±8.0	12.3±6.8	<0.001

*P values for the comparisons of WHO FC and PAH classification were calculated using Fisher exact test. 6MWD indicates 6-minute walking distance; CI, cardiac index; DBP, diastolic blood pressure; I/HPAH, idiopathic/heritable pulmonary arterial hypertension; LV GLS, left ventricular global longitudinal strain; LV SVi, left ventricular stroke volume index; LVEDVi, left ventricular end-diastolic volume index; LVEF, left ventricular ejection fraction; LVESVi, left ventricular end-systolic volume index; mPAP, mean pulmonary artery pressure; mRAP, mean right atrial pressure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PAH, pulmonary arterial hypertension; PAH-CHD, congenital heart disease-associated pulmonary arterial hypertension; PAH-CTD, connective tissue disease-associated pulmonary arterial hypertension; PCWP, pulmonary capillary wedge pressure; PVR, pulmonary vascular resistance; RA, right atrial; RVEDVi, right ventricular end-diastolic volume index; RVEF, right ventricular ejection fraction; RVESVi, right ventricular end-systolic volume index; RV FWLS, right ventricular free-wall longitudinal strain; RV GLS, right ventricular global longitudinal strain; SBP, systolic blood pressure; SvO₂, mixed venous oxygen saturation; and WHO FC, World Health Organization functional class.

CMR Analysis

Compared with survivors, nonsurvivors had significantly worse biventricular structure and function. Specifically, RV end-diastolic volume index and RVESVi were significantly higher in the deceased group (139±64 versus 108±47 mL/m²; $P<0.001$; and 98±50 versus 67±40 mL/m²; $P<0.001$, respectively), whereas RVEF was significantly lower (31±12% versus 41±15%; $P<0.001$). Left ventricular end-diastolic volume index ($P=0.41$), left ventricular end-systolic volume index ($P=0.11$), and left ventricular SVi ($P=0.06$) showed no significant differences between survivors and nonsurvivors, except for a slightly lower LV ejection fraction in the deceased group (57±12% versus 60±12%; $P=0.02$).

With respect to strain parameters, nonsurvivors demonstrated significantly impaired LV GLS (−17.0±5.2% versus −20.7±3.7%; $P<0.001$), RV GLS (−15.8±6.9% versus −20.3±7.5%; $P<0.001$), and RV FWLS (−16.6±8.3% versus −22.3±8.6%; $P<0.001$). Similarly, all 3 components of RA strain were significantly lower in the deceased group, including reservoir strain (20.5±9.9% versus 30.1±11.5%; $P<0.001$), conduit strain (7.6±4.6% versus 11.7±7.4%; $P<0.001$), and booster strain (12.3±6.8% versus 17.8±8.0%; $P<0.001$). More details are presented in Table 1.

Follow-Up and Outcomes

With a median follow-up duration of 39.3 months (interquartile range, 19.6–65.3 months), 68 patients reached the primary end point, and 91 patients reached the secondary end point.

Feature Selection

A total of 15 candidate variables were included in the elastic net Cox regression model, encompassing clinical characteristics, hemodynamic parameters, right- and left-ventricular functional indices, and strain measures.

As shown in Figure S3, for the primary end point (panel A), 5 variables were retained in the final model, including ln(NT-proBNP; standardized coefficient: 0.56), LV GLS (0.22), RA reservoir strain (−0.09), WHO FC (0.29), and RVEF (−0.09). For the composite end point (panel B), 4 parameters were finally selected, including ln(NT-proBNP; standardized coefficient, 0.73), WHO FC (0.30), LV GLS (0.22), and RA reservoir strain (−0.27).

To further compare the prognostic value of RA reservoir strain against RV GLS and RV FWLS, stepwise Cox proportional hazards regression based on the AIC was performed. Using the LV GLS-only model as the baseline adding RA reservoir strain resulted in the greatest improvement in model fit ($\Delta\text{AIC}=-13.1$). In contrast, adding RV GLS ($\Delta\text{AIC}=1.8$) or RV FWLS ($\Delta\text{AIC}=0.8$) yielded minimal incremental value. When RA reservoir strain was already included in the model with LV GLS, the further addition of RV GLS or RV FWLS did not meaningfully improve model performance. More details are presented in Table 2.

Survival Analysis

In univariate Cox analysis, each 5% increase in LV GLS (hazard ratio [HR], 2.23 [95% CI, 1.72–2.82]; $P<0.001$) was associated with an increased risk of the primary

Table 2. Comparison of Model Performance Using AIC for Different Strain Parameter Combinations

Model	AIC	Delta AIC
LV GLS	585.0	0.0
LV GLS+RA reservoir strain	572.0	−13.1
LV GLS+RV GLS	586.8	1.8
LV GLS+RV FWLS	585.9	0.8
LV GLS+RA reservoir strain+RV GLS	572.6	−12.5
LV GLS+RA reservoir strain+RV FWLS	573.7	−11.3

AIC indicates Akaike information criterion; LV GLS, left ventricular global longitudinal strain; RA, right atrial; RV FWLS, right ventricular free-wall longitudinal strain; and RV GLS, right ventricular global longitudinal strain.

end point, whereas each 5% increase in RA reservoir strain (HR, 0.65 [95% CI, 0.57–0.75]; $P<0.001$) was associated with a lower risk. After adjustment for WHO FC, ln (NT-proBNP), and RVEF, both LV GLS (per 5% increase, HR, 1.46 [95% CI, 1.07–2.00]; $P=0.02$) and RA reservoir strain (per 5% increase, HR, 0.77 [95% CI, 0.62–0.95]; $P=0.02$) remained independent predictors. For the composite end point, each 5% increase in LV GLS (HR, 2.04 [95% CI, 1.65–2.51]; $P<0.001$) was associated with an increased risk, whereas each 5% increase in RA reservoir strain (HR, 0.72 [95% CI, 0.65–0.81]; $P<0.001$) was associated with a lower risk. After adjustment for WHO FC, 6-minute walking distance, ln(NT-proBNP), and RVEF, both LV GLS (per 5% increase, HR, 1.44 [95% CI, 1.03–2.00]; $P=0.03$) and RA reservoir strain (per 5% increase, HR, 0.74 [95% CI, 0.60–0.92]; $P=0.007$) remained independent predictors. More details are presented in Tables 3 and 4.

Patients were stratified into 3 groups based on RVEF cutoffs of 19% and 45%. For the primary end point, both RA reservoir strain and LV GLS remained significant predictors across all RVEF subgroups, with a 1% decrease in the absolute value of either strain parameter associated with an increased risk of the adverse event. In patients with PAH and preserved RVEF (RVEF $>45\%$), RA reservoir strain and LV GLS were also prognostic factors for both end points. More details are presented in Figure 1.

Combination of Strain Parameters for Prognostication in Patients With PAH

We evaluated the incremental prognostic value of strain parameters beyond clinical variables and traditional CMR measures. For the primary endpoint, the combination of LV GLS and RA reservoir strain yielded significantly better performance than the ESC/ERS CMR model comprising RVEF, RVESVi, and SVi (χ^2 value: 50.7 versus 38.8; $P<0.001$). The C-index increased from 0.77 (95% CI, 0.70–0.84) to 0.84 (95% CI, 0.78–0.90; $P=0.05$). Adding LV GLS and RA reservoir strain improved the performance of the clinical model comprising WHO FC, 6-minute walking distance, NT-proBNP, and RVEF (χ^2 value: 92.6 versus 77.3; $P<0.001$), with the C-index increasing from 0.81 (95% CI, 0.75–0.87) to 0.86 (95% CI, 0.81–0.92; $P=0.008$). For the composite endpoint, LV GLS combined with RA reservoir strain outperformed the ESC/ERS model (χ^2 : 49.2 versus 40.5; $P=0.003$), with the C-index increasing from 0.75 (95% CI, 0.67–0.82) to 0.84 (95% CI, 0.78–0.89; $P=0.005$). Adding the strain parameters to the clinical model further improved its performance (χ^2 : 102.9 versus 69.1; $P<0.001$), with the C-index increasing from 0.76 (95% CI, 0.69–0.82) to 0.82 (95% CI, 0.77–0.87; $P=0.002$). More details are presented in Figure 2.

Using the Youden index, optimal cutoff values were identified for LV GLS (-18.2%) and RA reservoir strain

Table 3. Univariate Cox Analysis for the Primary and the Composite End Point

	Primary end point		Composite end point	
	HR (95% CI)	P value	HR (95% CI)	P value
Male	1.60 (0.93–2.74)	0.09	1.14 (0.69–1.87)	0.61
WHO FC	2.11 (1.48–2.99)	<0.001	1.92 (1.43–2.59)	<0.001
6MWD, m, per 25 increase	0.89 (0.85–0.93)	<0.001	0.94 (0.90–0.98)	0.003
ln(NT-proBNP), ng/L	2.30 (1.84–2.89)	<0.001	1.83 (1.54–2.17)	<0.001
LVEF, %, per 10 increase	0.77 (0.63–0.95)	0.01	0.69 (0.59–0.81)	<0.001
LV SVi, mL/m ² , per 10 increase	0.77 (0.62–0.96)	0.02	0.76 (0.63–0.93)	0.006
RVESVi, mL/m ² , per 10 increase	1.13 (1.09–1.18)	<0.001	1.10 (1.06–1.14)	<0.001
RVEDVi, mL/m ² , per 10 increase	1.08 (1.05–1.12)	<0.001	1.06 (1.03–1.09)	<0.001
RVEF, %, per 10 increase	0.65 (0.55–0.79)	<0.001	0.63 (0.54–0.74)	<0.001
LV GLS, %, per 5 increase	2.23 (1.75–2.82)	<0.001	2.04 (1.65–2.51)	<0.001
Impaired LV GLS	2.79 (1.72–4.53)	<0.001	2.36 (1.56–3.57)	<0.001
RV GLS, %, per 5 increase	1.48 (1.24–1.77)	<0.001	1.46 (1.26–1.69)	<0.001
RV FWLS, %, per 5 increase	1.39 (1.20–1.61)	<0.001	1.40 (1.23–1.58)	<0.001
RA reservoir strain, %, per 5 increase	0.65 (0.57–0.75)	<0.001	0.72 (0.65–0.81)	<0.001
Impaired RA reservoir strain	4.49 (2.79–7.23)	<0.001	3.34 (2.19–5.09)	<0.001
RA conduit strain, %, per 5 increase	0.59 (0.46–0.76)	<0.001	0.63 (0.52–0.77)	<0.001
RA booster strain, %, per 5 increase	0.64 (0.54–0.76)	<0.001	0.71 (0.62–0.81)	<0.001

6MWD indicates 6-minute walking distance; HR, hazard ratio; LVEF, left ventricular ejection fraction; LV GLS, left ventricular global longitudinal strain; LV SVi, left ventricular stroke volume index; NT-proBNP, N-terminal pro-B-type natriuretic peptide; RA, right atrial; RVEDVi, right ventricular end-diastolic volume index; RVEF, right ventricular ejection fraction; RVESVi, right ventricular end-systolic volume index; RV FWLS, right ventricular free-wall longitudinal strain; RV GLS, right ventricular global longitudinal strain; and WHO FC, World Health Organization functional class.

Table 4. Multivariate Cox Analysis for the Primary and the Composite End Point

	Primary end point		Composite end point	
	HR (95% CI)	P value	HR (95% CI)	P value
WHO FC	1.48 (1.02–2.14)	0.04	1.66 (1.15–2.40)	0.007
ln(NT-proBNP), ng/L	1.82 (1.39–2.37)	<0.001	1.56 (1.20–2.03)	<0.001
RVEF, %, per 10 increase	1.24 (0.97–1.58)	0.08	1.06 (0.82–1.38)	0.65
LV GLS, %, per 5 increase	1.46 (1.07–2.00)	0.02	1.44 (1.03–2.00)	0.03
RA reservoir strain, %, per 5 increase	0.77 (0.62–0.95)	0.02	0.74 (0.60–0.92)	0.007

HR indicates hazard ratio; LV GLS, left ventricular global longitudinal strain; NT-proBNP, N-terminal pro-B-type natriuretic peptide; RA, right atrial; and WHO FC, World Health Organization functional class.

(18.4%). Based on these thresholds, a risk heatmap was constructed (Figure S4). Patients with LV GLS <−18.2% and RA reservoir strain >18.4% were classified as low risk, those with LV GLS >−18.2% and RA reservoir strain <18.4% as high risk, and the remaining patients as intermediate risk. For the primary end point, the cumulative event probability was significantly different among all 3 groups. The low-risk group showed significantly lower event rates than the intermediate-risk group ($P<0.001$) and the high-risk group ($P<0.001$), whereas the intermediate-risk group also had a lower risk than the high-risk group ($P=0.002$). Similarly, for the composite end point, the low-risk group exhibited the most favorable prognosis, with significantly lower event probabilities than both the intermediate-risk group ($P=0.001$) and the high-risk group ($P<0.001$).

The difference between the intermediate- and high-risk groups also reached statistical significance ($P=0.001$). More details are presented in Figure 3.

One-Year Mortality and Long-term Prognosis According to Strain Risk Groups

One-year event risk across strain-defined risk groups was presented in Figure 4. One-year mortality rates were 2.8% (95% CI, 0.1%–5.5%), 6.2% (95% CI, 0.9%–11.5%), and 21.0% (95% CI, 8.1%–33.9%) in the low-, intermediate-, and high-risk groups, respectively. Notably, this classification aligns well with the 2022 ESC/ERS guideline-defined risk categories, effectively stratifying patients into low-, intermediate-, and high-risk groups.

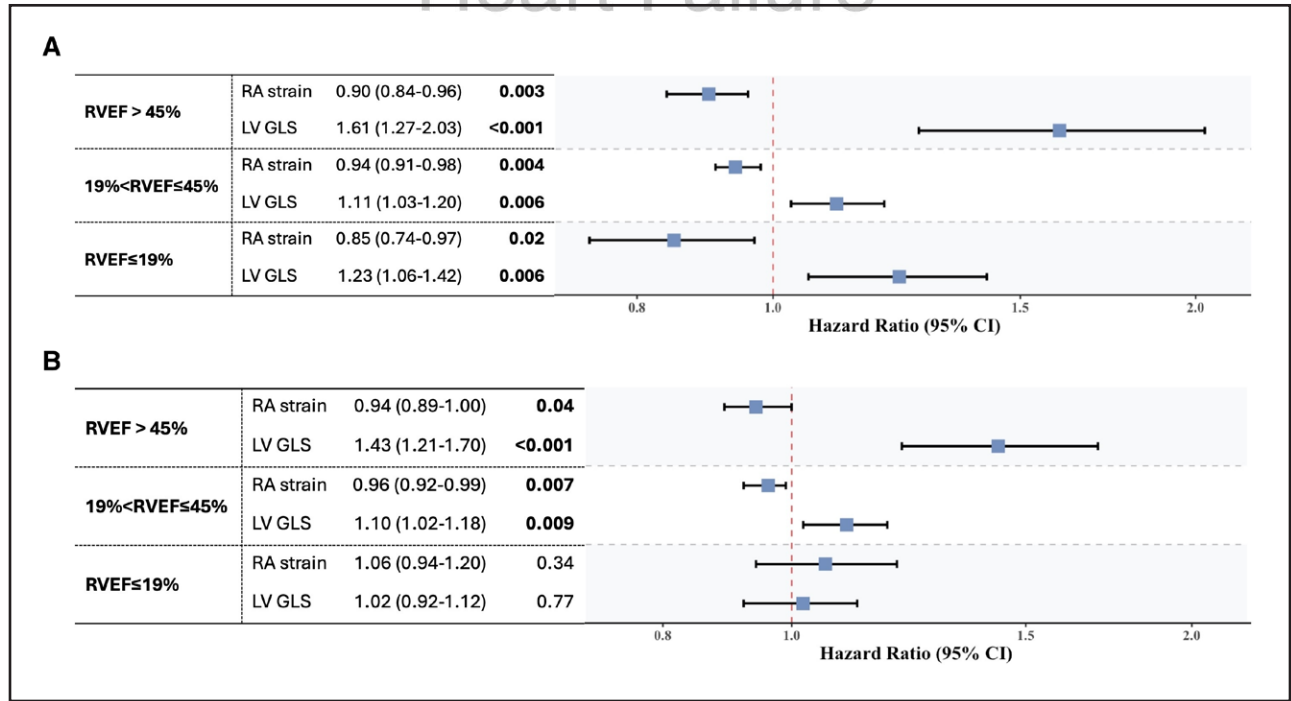


Figure 1. Subgroup analysis of the prognostic value of right atrial (RA) reservoir strain and left ventricular global longitudinal strain (LV GLS) based on right ventricular ejection fraction (RVEF)-defined categories. Forest plots for the primary end point (A) and secondary end point (B) stratified by RVEF using 19% and 45% as cutoff values. The prognostic impact of RA reservoir strain and LV GLS was evaluated within each RVEF subgroup. The 3 panels (from top to bottom) correspond to patients with RVEF >45%, 19%<RVEF≤45%, and RVEF ≤19%, respectively.

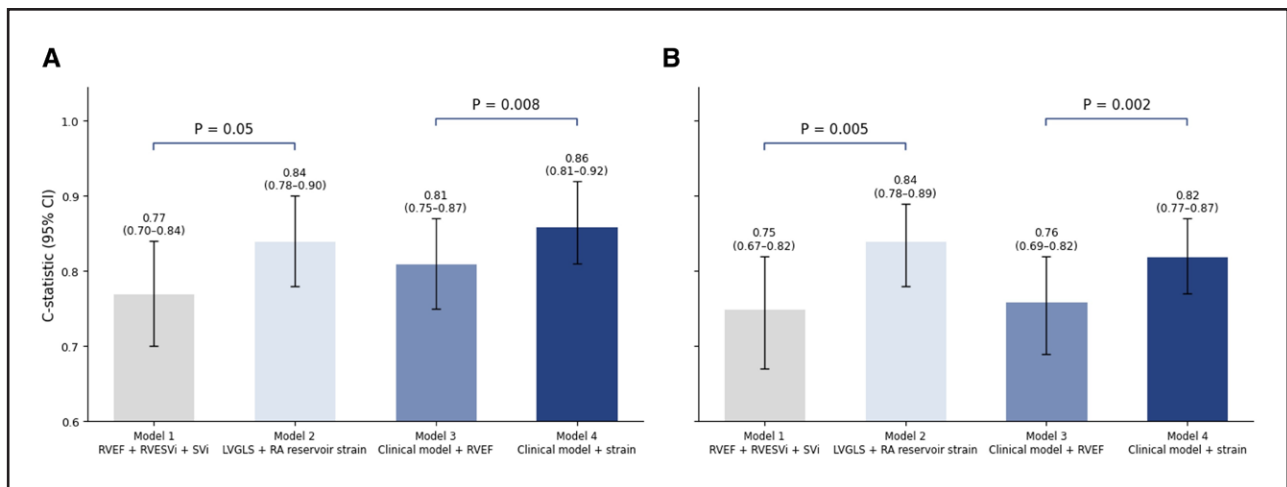


Figure 2. Incremental prognostic value of different models for predicting primary and composite end points.

Bar plots display the C statistics of 4 prognostic models for the primary end point (A) and composite end point (B). Model 1 includes traditional cardiac magnetic resonance (CMR) parameters (stroke volume index [SVi], right ventricular end-systolic volume index [RVESVi], and right ventricular ejection fraction [RVEF]); model 2 includes left ventricular global longitudinal strain (LV GLS) and right atrial (RA) reservoir strain; model 3 includes the noninvasive clinical and conventional CMR-derived RV parameters (World Health Organization functional class [WHO FC], 6-minute walking distance [6MWD], NT-proBNP [N-terminal pro-B-type natriuretic peptide], and RVEF); and model 4 combines the noninvasive clinical parameters with LV GLS and RA reservoir strain (WHO FC, 6MWD, NT-proBNP, LV GLS, and RA reservoir strain).

To assess the long-term prognostic value of strain-based risk stratification, a landmark analysis was performed among patients who survived >1 year. Cumulative probability of death from the 1-year

landmark differed significantly among the strain groups, with a higher risk observed in the intermediate- and high-risk groups compared with the low-risk group (low versus intermediate: $P=0.01$; low versus

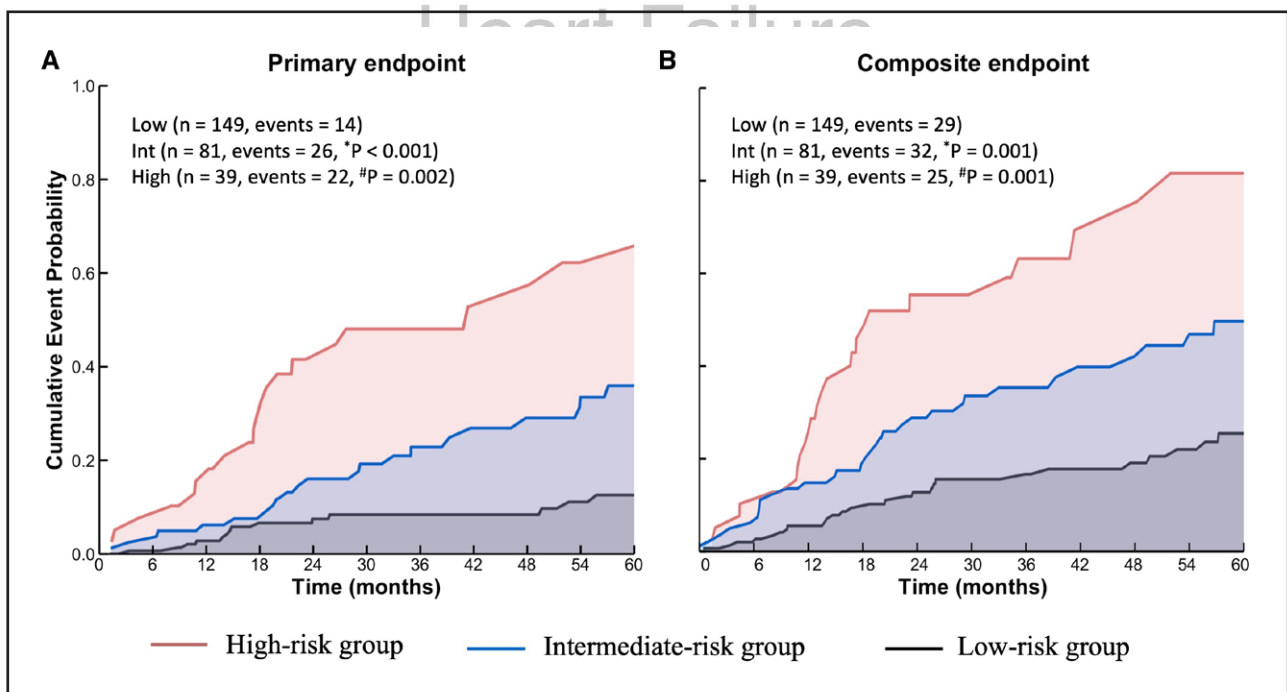


Figure 3. Cumulative event probability curves stratified by left ventricular global longitudinal strain (LV GLS) and right atrial (RA) reservoir strain-based risk groups.

Cumulative event probabilities are shown for the primary end point (A) and composite end point (B). Patients were categorized as low-, intermediate-, and high-risk groups based on LV GLS and RA reservoir strain. The low-risk group included patients with both normal LV GLS and RA reservoir strain, whereas the high-risk group included those with impaired LV GLS and RA reservoir strain; the remaining patients were classified as intermediate risk. * P denotes differences between the low-risk group and the intermediate-risk group. # P denotes differences between the intermediate-risk group and the high-risk group.

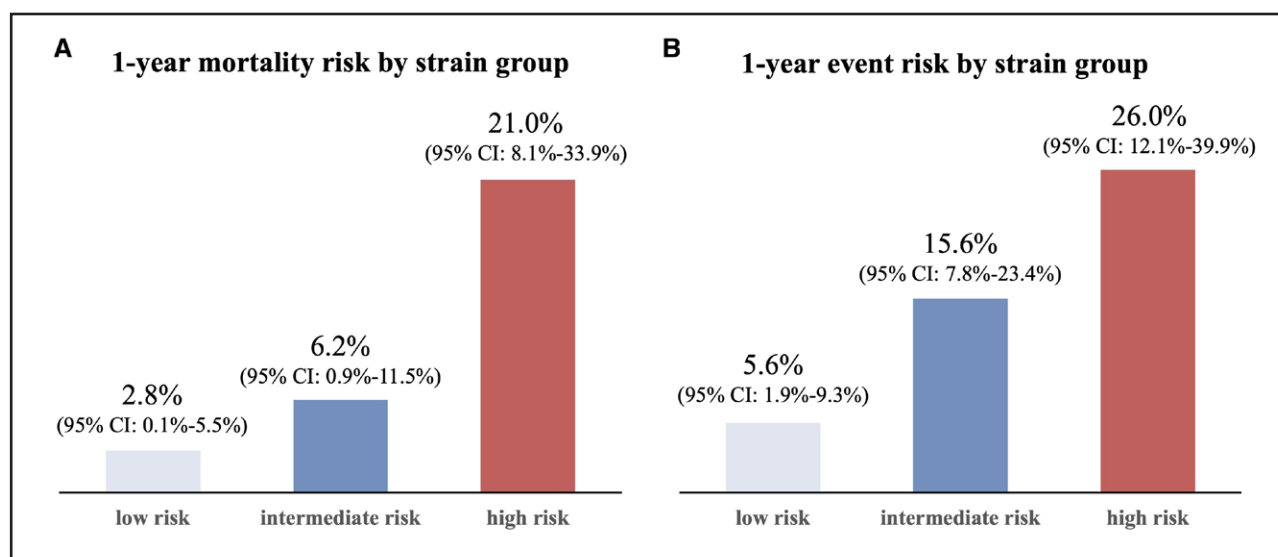


Figure 4. One-year mortality and event risk according to strain-based risk groups.

Bar charts display the 1-year mortality risk (A) and 1-year composite event risk (B) in patients with pulmonary arterial hypertension (PAH) stratified by left ventricular global longitudinal strain (LV GLS) and right atrial (RA) reservoir strain.

high: $P < 0.001$), whereas the difference between the intermediate- and high-risk groups was not statistically significant ($P = 0.06$). More details are presented in Figure S5.

DISCUSSION

The main findings of the study include: (1) of all strain parameters, the combination of LV GLS and RA reservoir strain provided the highest prognostic value for outcome prediction in patients with PAH and preserved its discriminatory capacity during both short- and long-term follow-up; (2) both LV GLS and RA reservoir strain emerged as independent predictors of all-cause mortality and composite clinical events, even after adjustment for WHO FC, 6-minute walk distance, and RVEF; and (3) the integration of LV GLS and RA reservoir strain significantly outperformed the current ESC/ERS-recommended CMR model and improved the predictive performance of the traditional clinical model.

Recent work has strengthened the importance of strain analysis in prognostication among patients with PAH. RV longitudinal strain, particularly RV FWLS, emerged as a strong imaging predictor of death and composite events.^{7,8,20} LV systolic function and LV GLS correlated with RV dysfunction, and LV GLS $> -14.2\%$ was associated with poor clinical outcomes in precapillary patients with PH.²¹ Emerging data showed RA reservoir strain deteriorates earlier than RV systolic function and independently predicts mortality, underscoring the pivotal role of atrial-ventricular interaction in PAH.^{22,23} Similar findings were observed in this study. All strain parameters demonstrated prognostic value for both the composite end point and all-cause mortality in this study,

with RA reservoir strain and LV GLS emerging as independent predictors.

Notably, in this study, the combination of LV GLS and RA reservoir strain emerged as the most effective prognostic strain combination, whereas RV strain parameters were not retained. We first applied the elastic net approach to select the most effective clinical and imaging predictors of outcome, followed by stepwise AIC analysis to further examine why RV strain indices were excluded from the final model. Their exclusion did not imply a lack of prognostic value, but rather inferior predictive performance compared with RA reservoir strain. As a result, the model selected the combination of LV GLS and RA reservoir strain, which provided the most comprehensive prognostic information. Importantly, LV GLS may contribute complementary prognostic information through ventricular interdependence. In PAH, progressive RV pressure overload leads to interventricular septal displacement and impaired LV filling, resulting in secondary alterations in LV longitudinal function. Therefore, LV GLS may reflect the systemic consequences of right heart remodeling and impaired biventricular coupling, which may explain why it retained prognostic value even in patients with more advanced disease and reduced RVEF. Awareness of the RA-RV interaction in PAH is increasing.¹¹ An increasing number of studies have demonstrated the direct impact of RV diastolic stiffness on RA function.^{12,24} Chronic RV pressure overload alters RA size and function, which are linked to symptoms and prognosis.²⁵ Backward flow from functional tricuspid regurgitation further amplifies RA remodeling and is recognized as a key pathophysiologic mechanism.²⁶ Moreover, recent evidence indicates that significant impairment of RA phasic function often emerges early in the disease course of PH before the



RV becomes overloaded and begins to fail.²² In PAH, LV changes were characterized mainly by underfilling due to elevated RV pressure and ventricular interdependence, rather than primary LV disease.²⁷ Chronic RV pressure overload shifted the septum leftward within the pericardial constraint, reducing LV compliance and end-diastolic filling. Recent CMR studies further demonstrated that LV dysfunction in PAH was characterized by smaller LV and left atrial volumes and impaired pulmonary venous return, underscoring underfilling as the key mechanism.^{16,28} These changes are accompanied by reductions in longitudinal shortening and torsion, even when global ejection fraction is preserved.²⁹ In our study, we observed that alterations in RA reservoir strain and LV GLS often preceded the decline in RVEF. Using 19% and 45% as cutoff values of RVEF for subgroup analyses, we found that both LV GLS and RA reservoir strain remained significantly associated with adverse outcomes even in patients with preserved RVEF. These findings further support that LV GLS and RA reservoir strain can capture early changes in right heart function in PAH before RV systolic impairment.

Identifying high-risk patients remains a major challenge in PAH management.^{30,31} The 2022 ESC/ERS guidelines incorporated CMR parameters, including RVEF, RVESVi, and SVi, for risk assessment in patients with PAH.³ However, these volumetric indicators do not adequately reflect early myocardial dysfunction, diastolic impairment, or RA remodeling, which are now increasingly recognized as key determinants of adverse outcomes in PAH.^{8,22,29} The Seventh World Symposium on Pulmonary Hypertension also emphasized the need for further evaluation of emerging CMR techniques, such as feature-tracking strain analysis, late gadolinium enhancement, 4-dimensional flow imaging, T_1/T_2 mapping, and RA and PA parameters, to better define their role in risk stratification and outcome prediction in PAH.³² Thus, in this study, we demonstrated that LV GLS and RA reservoir strain outperformed conventional CMR parameters for prognostic evaluation and effectively stratified patients with PAH into different risk categories, underscoring their potential value for risk stratification. Moreover, among patients who survived >1 year, the combined strain parameters remained predictive of long-term outcomes, emphasizing their consistent prognostic utility for both short- and long-term outcome prediction.

However, our study has several limitations. First, strain analysis was based on CMR, which may not be applicable to patients with PAH who are unable to undergo CMR due to contraindications, severe disease instability, or difficulties with breath-holding. Whether echocardiography-derived LV GLS provides similar prognostic value in PAH warrants further investigation. Second, only a subset of patients had 1-year follow-up CMR examinations, limiting our ability to comprehensively assess the prognostic significance of longitudinal

changes in strain parameters. Third, we did not specifically assess the incremental value of LV GLS and RA reservoir strain beyond established PAH risk models, such as REVEAL, ESC/ERS, COMPERA, or French noninvasive models. Further studies are warranted to determine whether LV GLS and RA reservoir strain can improve risk prediction when added to these validated risk scores.

CONCLUSIONS

The combination of LV GLS and RA reservoir strain provides the strongest prognostic value in PAH and offers incremental improvement beyond current clinical and imaging models, supporting its potential integration into refined risk stratification strategies.

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Disclosures

None.

Supplemental Material

Supplemental Methods
Figures S1–S5

REFERENCES

- Kovacs G, Bartolome S, Denton CP, Gatzoulis MA, Gu S, Khanna D, Badesch D, Montani D. Definition, classification and diagnosis of pulmonary hypertension. *Eur Respir J*. 2024;64:2401324. doi: 10.1183/13993003.01324-2024
- GBD 2021 Pulmonary Arterial Hypertension Collaborators. Global, regional, and national burden of pulmonary arterial hypertension, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *Lancet Respir Med*. 2025;13:69–79. doi: 10.1016/S2213-2600(24)00295-9
- Humbert M, Kovacs G, Hoeper MM, Badagliacca R, Berger RMF, Bida M, Carlsen J, Coats AJS, Escribano-Subias P, Ferrari P, et al; ESC/ERS Scientific Document Group. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Heart J*. 2022;43:3618–3731. doi: 10.1093/eurheartj/ehac237
- Alabed S, Shahin Y, Garg P, Alandejani F, Johns CS, Lewis RA, Condliffe R, Wild JM, Kiely DG, Swift AJ. Cardiac-MRI predicts clinical worsening and mortality in pulmonary arterial hypertension: a systematic review and meta-analysis. *JACC Cardiovasc Imaging*. 2021;14:931–942. doi: 10.1016/j.jcmg.2020.08.013
- Jayasekera G, Macdonald A, McComb C, Orchard V, Welsh D, Church C, Johnson M, Brewis M, Berry C, Radjenovic A, et al. Left ventricular dysfunction and intra-ventricular dyssynchrony in idiopathic

- pulmonary arterial hypertension. *Int J Cardiol.* 2022;365:131–139. doi: 10.1016/j.ijcard.2022.07.032
6. Haecck ML, Scherptong RW, Marsan NA, Holman ER, Schalij MJ, Bax JJ, Vliegen HW, Delgado V. Prognostic value of right ventricular longitudinal peak systolic strain in patients with pulmonary hypertension. *Circ Cardiovasc Imaging.* 2012;5:628–636. doi: 10.1161/CIRCIMAGING.111.971465
 7. de Siqueira ME, Pozo E, Fernandes VR, Sengupta PP, Modesto K, Gupta SS, Barbeito-Caamano C, Narula J, Fuster V, Caixeta A, et al. Characterization and clinical significance of right ventricular mechanics in pulmonary hypertension evaluated with cardiovascular magnetic resonance feature tracking. *J Cardiovasc Magn Reson.* 2016;18:39. doi: 10.1186/s12968-016-0258-x
 8. Nabeshima Y, Kitano T, Node K, Takeuchi M. Prognostic value of right ventricular free-wall longitudinal strain in patients with pulmonary hypertension: systematic review and meta-analyses. *Open Heart.* 2024;11:e002561. doi: 10.1136/openhrt-2023-002561
 9. Hahn RT, Lerakis S, Delgado V, Addetia K, Burkhoff D, Muraru D, Pinney S, Friedberg MK. Multimodality imaging of right heart function: JACC scientific statement. *J Am Coll Cardiol.* 2023;81:1954–1973. doi: 10.1016/j.jacc.2023.03.392
 10. Leng S, Dong Y, Wu Y, Zhao X, Ruan W, Zhang G, Allen JC, Koh AS, Tan RS, Yip JW, et al. Impaired cardiovascular magnetic resonance-derived rapid semiautomated right atrial longitudinal strain is associated with decompensated hemodynamics in pulmonary arterial hypertension. *Circ Cardiovasc Imaging.* 2019;12:e008582. doi: 10.1161/CIRCIMAGING.118.008582
 11. Richter MJ, Fortuni F, Alenezi F, D'Alto M, Badagliacca R, Brunner NW, van Dijk AP, Douschan P, Gall H, Ghio S, et al. Imaging the right atrium in pulmonary hypertension: a systematic review and meta-analysis. *J Heart Lung Transpl.* 2023;42:433–446. doi: 10.1016/j.healun.2022.11.007
 12. Wessels JN, van Wezenbeek J, de Rover J, Smal R, Lluçà-Valleperas A, Celant LR, Marcus JT, Meijboom LJ, Groeneveldt JA, Oosterveer FPT, et al. Right atrial adaptation to precapillary pulmonary hypertension: pressure-volume, cardiomyocyte, and histological analysis. *J Am Coll Cardiol.* 2023;82:704–717. doi: 10.1016/j.jacc.2023.05.063
 13. Galiè N, Humbert M, Vachiery JL, Gibbs S, Lang I, Torbicki A, Simonneau G, Peacock A, Vonk Noordegraaf A, Beghetti M, et al; ESC Scientific Document Group. 2015 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension: The Joint Task Force for the Diagnosis and Treatment of Pulmonary Hypertension of the European Society of Cardiology (ESC) and the European Respiratory Society (ERS); endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC), International Society for Heart and Lung Transplantation (ISHLT). *Eur Heart J.* 2016;37:67–119. doi: 10.1093/eurheartj/ehv317
 14. Callan P, Clark AL. Right heart catheterisation: indications and interpretation. *Heart.* 2016;102:147–157. doi: 10.1136/heartjnl-2015-307786
 15. Kramer CM, Barkhausen J, Bucciarelli-Ducci C, Flamm SD, Kim RJ, Nagel E. Standardized cardiovascular magnetic resonance imaging (CMR) protocols: 2020 update. *J Cardiovasc Magn Reson.* 2020;22:17. doi: 10.1186/s12968-020-00607-1
 16. Guo J, Wang J, Wang L, Li Y, Xu Y, Li W, Chen C, He J, Yin L, Pu S, et al. Left ventricular underfilling in PAH: a potential indicator for adaptive-to-maladaptive transition. *Pulm Circ.* 2023;13:e12309. doi: 10.1002/pul.12309
 17. Schulz-Menger J, Bluemke DA, Bremerich J, Flamm SD, Fogel MA, Friedrich MG, Kim RJ, von Knobelsdorff-Brenkenhoff F, Kramer CM, Pennell DJ, et al. Standardized image interpretation and post-processing in cardiovascular magnetic resonance—2020 update: Society for Cardiovascular Magnetic Resonance (SCMR): Board of Trustees Task Force on Standardized Post-Processing. *J Cardiovasc Magn Reson.* 2020;22:19. doi: 10.1186/s12968-020-00610-6
 18. Celant LR, Wessels JN, Marcus JT, Meijboom LJ, Bogaard HJ, de Man FS, Vonk Noordegraaf A. Towards the implementation of optimal cardiac magnetic resonance risk stratification in PAH. *Chest.* 2024;165:181–191. doi: 10.1016/j.chest.2023.07.028
 19. Simon N, Friedman J, Hastie T, Tibshirani R. Regularization paths for Cox's proportional hazards model via coordinate descent. *J Stat Softw.* 2011;39:1–13. doi: 10.18637/jss.v039.i05
 20. Hulshof HG, Eijssvogels TMH, Kleinnibbelink G, van Dijk AP, George KP, Oxborough DL, Thijssen DHJ. Prognostic value of right ventricular longitudinal strain in patients with pulmonary hypertension: a systematic review and meta-analysis. *Eur Heart J Cardiovasc Imaging.* 2019;20:475–484. doi: 10.1093/ehjci/jej120
 21. Padervinskienė L, Krivickienė A, Hoppenot D, Miliauskas S, Basevičius A, Nedzelskienė I, Jankauskas A, Šimkus P, Ereminienė E. Prognostic value of left ventricular function and mechanics in pulmonary hypertension: a pilot cardiovascular magnetic resonance feature tracking study. *Medicina.* 2019;55:73. doi: 10.3390/medicina55030073
 22. Yang F, Yan Y, Jiang W, Wang Z, Wu C, Wu Q, Deng Y, Du Y, Yang Z, Zhang Z, et al. Impaired right atrial function preceding right ventricular systolic dysfunction: clinical utility and long-term prognostic value in pulmonary hypertension. *Insights Imaging.* 2025;16:115. doi: 10.1186/s13244-025-01996-6
 23. Vos JL, Leiner T, van Dijk APJ, van der Zwaan HB, Sieswerda GT, Snijder RJ, Post MC, Vonk MC, van Leuven S, Vart P, et al. Right atrial and ventricular strain detects subclinical changes in right ventricular function in precapillary pulmonary hypertension. *Int J Cardiovasc Imaging.* 2022;38:1699–1710. doi: 10.1007/s10554-022-02555-6
 24. Costanzo MR, Marwick TH. Right atrium as barometer of right ventricular stiffness and treatment responsiveness in precapillary pulmonary hypertension. *J Am Coll Cardiol.* 2023;82:718–720. doi: 10.1016/j.jacc.2023.06.009
 25. Alenezi F, Mandawat A, Il'Giovine ZJ, Shaw LK, Siddiqui I, Tapson VF, Arges K, Rivera D, Romano MMD, Velazquez EJ, et al. Clinical utility and prognostic value of right atrial function in pulmonary hypertension. *Circ Cardiovasc Imaging.* 2018;11:e006984. doi: 10.1161/CIRCIMAGING.117.006984
 26. Yoshida K, van Wezenbeek J, Wessels JN, de Man FS, Sunagawa K, Vonk-Noordegraaf A, Bogaard HJ. Tricuspid regurgitation in pulmonary arterial hypertension: a right ventricular volumetric and functional analysis. *Eur Respir J.* 2024;63:2301696. doi: 10.1183/13993003.01696-2023
 27. Venkateshvaran A, Bohlin J, Kjellström B, Bergström E, Nelsson A, Werther Evaldsson A, Rådegran G, Arheden H, Ostenfeld E. Left ventricular dysfunction in pulmonary arterial hypertension is attributed to underfilling rather than intrinsic myocardial disease: a CMR 2D phase contrast study. *Sci Rep.* 2024;14:17280. doi: 10.1038/s41598-024-68254-5
 28. Naeije R, Badagliacca R. The overloaded right heart and ventricular interdependence. *Cardiovasc Res.* 2017;113:1474–1485. doi: 10.1093/cvr/cvx160
 29. Sjögren H, Kjellström B, Bredfeldt A, Steding-Ehrenborg K, Rådegran G, Hesselstrand R, Arheden H, Ostenfeld E. Underfilling decreases left ventricular function in pulmonary arterial hypertension. *Int J Cardiovasc Imaging.* 2021;37:1745–1755. doi: 10.1007/s10554-020-02143-6
 30. Ford HJ, LeVarge B. The pursuit of risk assessment and stratification tools in pulmonary arterial hypertension: have we seen the lite? *Chest.* 2021;159:14–16. doi: 10.1016/j.chest.2020.10.026
 31. Farber HW, Benza RL. Risk assessment tools in pulmonary arterial hypertension. Prognosis for prospective trials? *Am J Respir Crit Care Med.* 2018;197:843–845. doi: 10.1164/rccm.201801-0042ED
 32. Dardi F, Boucly J, Benza R, Frantz R, Mercurio V, Olschewski H, Rådegran G, Rubin LJ, Hoeper MM. Risk stratification and treatment goals in pulmonary arterial hypertension. *Eur Respir J.* 2024;64:2401323. doi: 10.1183/13993003.01323-2024